The Editor
Statistical Methods in Medical Research

Dear Editor,

We are pleased to submit our manuscript, "A Bayesian Joint Longitudinal–Interval and Multi-State Threshold-Crossing Framework for Irregular Molecular Monitoring in Chronic Myeloid Leukemia," for consideration as an original methodological development with application.

Routine molecular monitoring in chronic myeloid leukemia produces data that break the assumptions of standard survival and landmark analyses: log-transformed minimal residual disease is measured at irregular visits, is left-censored at an assay reporting floor, and the deep-molecular-response (DMR) endpoint is ascertained only at the visits when a patient is tested, and is defined by a threshold on the very biomarker used as its predictor. We develop a Bayesian framework built around two linked models that respect these features. The first couples a left-censored longitudinal sub-model with an interval-observed likelihood for first documented DMR through shared random effects. The second reads the endpoint directly as a latent threshold-crossing (first-hitting-time) process and extends it to a multi-state model of onset, durable response, and molecular relapse; because the trajectory is piecewise-quadratic, the running minimum and maximum that define these transitions are available in closed form, which removes the usual soft-minimum grid approximation and yields divergence-free Hamiltonian Monte Carlo.

We believe the manuscript suits the journal because it develops a general statistical methodology—latent threshold-crossing and multi-state joint modelling under assay-floor left-censoring and interval ascertainment—while motivating and illustrating it on a real clinical monitoring cohort, and is written to be accessible to a broad methodological and applied audience. A simulation study characterises identifiability, coverage, and the consequences of mishandling the assay floor; the application reports posterior predictive checks, calibration, model comparison, and dynamic prediction, with limitations (single-centre data, internal dynamic-prediction accuracy, and imperfect finite-sample coverage) stated plainly.

The manuscript is our original work, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and agree to its content. We have no conflicts of interest to declare. The manuscript has been anonymized for double-anonymized review, with author and institutional information provided on a separate Title Page.

We note that, with the addition of the multi-state model and the revisions requested at review, the manuscript's length (counting tables and figures) slightly exceeds the journal's guideline of around 8000 words; we are happy to move selected tables to the Supplementary Material at the editor's discretion.

Thank you for considering our work.

Sincerely,
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